

1. Administrative & Study Identification

Full Study Title: A Phase IIa Multicenter, Open-label, 12-week Study to Investigate the Pharmacokinetics and Safety and to Provide Proof of Mechanism of Alogabat in Children and Adolescents Aged 5-17 Years With Angelman Syndrome (AS) With Deletion Genotype **Short Title/Acronym:** Aldebaran Trial **Protocol Number:** BP41315 / 2022-501844-14-00 **NCT Number:** NCT05630066 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Alogabat **Phase of Development:** Phase IIa **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the pharmacokinetics (PK) and safety of alogabat in children and adolescents aged 5-17 years with Angelman Syndrome (AS) with a deletion genotype. **Secondary Objectives:** To evaluate the proof of mechanism by assessing changes in brain activity patterns (via electroencephalography/EEG) and to monitor adverse events over the 12-week study period.

2.2 Background & Rationale Angelman syndrome (AS) is a rare genetic disorder characterized by developmental delays, cognitive impairment, sleep disturbances, behavioral challenges, and frequent seizures. The most severe presentation of AS is often seen in individuals with the "deletion genotype," where genetic deletions on chromosome 15q11q13 affect not only the *UBE3A* gene but also neighboring genes that code for GABAA α 5 receptors. Alogabat is an experimental oral medication specifically designed to activate these remaining GABAA α 5 receptors to compensate for their reduced numbers, with the goal of addressing the core neurodevelopmental symptoms of the disease.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Open-label) **Blinding:** Open-label **Randomization:** N/A (Two-part study: Part 1 for dose confirmation; Part 2 for dose exploration based on Part 1 results)

3.2 Planned Enrollment & Duration Target \$N\$: Up to 56 pediatric participants **Number of Sites:** Multicenter (including sites in the US, Italy, Spain, and Australia) **Estimated Study Start:** Late 2022 / Early 2023 **Estimated Primary Completion:** ~24 weeks per participant (12 weeks of treatment + screening and follow-up)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 5 to ≤ 17 years. 2. Clinical diagnosis of AS and a genetic subtype of deletion on chromosome 15q11q13 confirmed by a historical molecular diagnosis. 3. The participant's general health status allows them to safely participate in the trial, as determined by the investigator.

4.2 Exclusion Criteria 1. Diagnosis of any type of Angelman syndrome other than the deletion genotype. 2. Hospitalization for any major medical or surgical procedure involving general anesthesia within 12 weeks of screening, or planned during the study. 3. Any prior or current treatment with an investigational study drug within 6 weeks (or 5 half-lives) prior to baseline, or concurrent uncontrolled cardiovascular disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Alogabat administered once daily (QD). The dosage is age-adjusted and cohort-dependent based on ongoing PK results from Part 1. **Route of Administration:** Oral tablet. **Duration of Treatment:** 12 weeks of daily administration.

5.2 Concomitant & Prohibited Therapy Permitted: Standard background care and permitted medications for seizure and behavioral management. **Prohibited:** Concurrent investigational drugs within 6 weeks prior to baseline, or medications known to interfere with alogabat metabolism.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Days -42 to -15	Eligibility assessment, historical molecular diagnosis review, baseline vitals
Baseline	Day 1	First Dose, Baseline PK, EEG, and Safety measures
Treatment Period	Weeks 2, 4, 12	Onsite Visits: PK sampling, EEG recordings (brain activity patterns), AE monitoring
Follow-up	Weeks 14 & 18	Post-treatment safety checks (2 and 6 weeks after completing treatment)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Age-group Based Ratio of Plasma PK Parameter: Area Under the Concentration-time Curve (AUC) from Day 1, Week 2, and Week 12. **Measurement Method:** Plasma pharmacokinetic blood sampling.

7.2 Secondary & Exploratory Endpoints * Change in brain activity patterns (measured by EEG beta-band power) from pre-baseline to Week 2, Week 4, and Week 12. * Incidence and severity of treatment-emergent adverse events (AEs) from baseline to Week 12.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits every 2-4 weeks during the 12-week treatment period, plus continuous caregiver reporting. **Serious Adverse Events (SAEs):** Reported promptly following standard GCP guidelines. **Data Monitoring Committee (DMC):** Internal safety data review, particularly between Part 1 (dose confirmation) and Part 2 (dose exploration) to ensure safe age-adjusted dosing limits.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of alogabac) and PK-evaluable population. **Sample Size Justification:** Up to 56 participants, determined to be sufficient for a Phase IIa PK, safety, and proof-of-mechanism evaluation in a rare pediatric population. **Handling of Missing Data:** Standard handling for early withdrawals, focusing on cumulative PK, EEG, and safety data emerging from all available timepoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and international regulatory standards. **Monitoring Plan:** Routine clinical site monitoring, safety assessments via continuous follow-up throughout the 24-week total trial period. **Audit Trail:** eCRF entry, data lock procedures, and protocol adherence checks at all participating global sites.

11. Study Identifiers & Governance

Primary ID: BP41315 **Registry Number:** NCT05630066 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** Aldebaran Trial **Official Regulatory Title:** A Phase IIa Multicenter, Open-label, 12-week Study to Investigate the Pharmacokinetics and Safety and to Provide Proof of Mechanism of Alogabac in Children and Adolescents Aged 5-17 Years With Angelman Syndrome (AS) With Deletion Genotype

12. Clinical Framework (Angelman Syndrome Specific)

Primary Condition: Angelman Syndrome (AS) **Diagnosis Threshold:** Confirmed genetic deletion on chromosome 15q11q13 **Medical Methodology:** Activation of targeted brain receptors (GABAA α 5 receptors) **Optimization Target:** Restoration of GABAA α 5 receptor function/signaling and improvement in EEG beta-band power

13. Study Procedures & Methodology

Management Pathway: Standard neurologic and developmental clinical assessment for Angelman Syndrome **Education Protocol (HCP):** Site initiation, protocol training, and precise EEG/PK sampling schedules **Education Protocol (Patient/Caregiver):** Caregiver instructions for once-daily oral tablet administration and adverse event diary tracking **Quality Audit Mechanism:** Source data verification, clinical site monitoring, and centralized EEG data reading

14. Observation & Follow-Up Schedule

Total Duration: ~24 weeks (up to 6 weeks screening + 12 weeks treatment + 6 weeks follow-up) **Onsite Visit Cadence:** Every 2-4 weeks during the active treatment period **Remote Monitoring Frequency:** Continuous caregiver check-ins as clinically indicated **Checklist Review Interval:** Routine safety evaluations at all scheduled visits

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				